From tan(x) to the Pond: A Conversation with Claude About Recurrence

A dialogue, lightly edited for typos, otherwise as it happened. My prompts are verbatim; Claude's replies are condensed from the originals with the key passages kept whole. Annotations mark where the conversation turned.

Context: this began as a review session for my local ML lab — a hand-built Go neural net alongside MLX LoRA fine-tuning — and a question about whether building differently creates models that think differently. The conclusion of that thread was that homogeneity lives in shared data and shared objectives, not in languages or frameworks. That set up the question that starts this dialogue: if the architecture itself is where I want differentiation, what's actually out there, and where does the map end?

Turn 1 — The lay of the land

Me: does a hybrid transformer and RNN exist?

Claude: Yes — active and increasingly mainstream. RWKV trains in parallel like a transformer but runs inference as an RNN, constant memory per token. Griffin/RecurrentGemma mix gated linear recurrences with local attention; Jamba and Zamba interleave Mamba layers — state-space models, recurrences in modern dress — with periodic full attention. The consistent finding: a mostly- recurrent stack with a *little* attention recovers nearly all of full- transformer quality at a fraction of the inference memory, because recurrence carries a compressed running state while occasional attention handles precise recall.

> **Turning point:** the reply ended by noting that recurrence is a *sensing* posture — a persistent internal state updated per input, rather than retrieve-the-whole-context. That's the frame the rest of the conversation runs on.

Turn 2 — First probe: break linearity

Me: why not polynomial recurrence?

Claude: It existed — 1990s second-order RNNs, Sutskever's multiplicative RNN — and lost on three fronts. Linear recurrence $h_t = A \cdot h_{t-1} + B \cdot x_t$ is associative, so the whole sequence unrolls with a parallel scan; that's the entire trick behind Mamba and RWKV training at transformer speed. Make the update polynomial in h and associativity dies — you're back to strictly sequential training. Second, iterated polynomials of degree ≥ 2 are the textbook generators of chaos (the logistic map is degree 2); positive Lyapunov exponents destroy gradient flow by definition. Third, a full bilinear state–state term is a $d \times d \times d$ tensor.

The sleight of hand: the field kept polynomial *interactions* but quarantined them where they can't compound. Mamba's selectivity is $A(x_t) \cdot h_{t-1}$ — bilinear in input×state but linear in state, so the scan survives. Gates are elementwise degree-2, bounded by sigmoids. Attention's query·key is itself degree-2 in the inputs. We use polynomial *coupling* everywhere but keep the *iterated* map linear, because iterating a nonlinear polynomial map is how you build a chaos generator, not a memory.

> **Turning point:** the distinction between *coupling* (allowed anywhere) and *iteration* (must stay linear). Every later move in the conversation respects this line without my having planned it.

Turn 3 — Second probe: motion without magnitude

Me: what if linear recurrence was more so tangential

Claude: If you mean recurrence whose update moves tangent to the state's current position — rotating rather than scaling — you've independently arrived at one of the elegant lines in this literature. A linear map with orthogonal or unitary A does exactly that: every step is motion tangent to the sphere of constant norm. Nothing grows, nothing shrinks; information is stored in *phase and direction*, not magnitude. That's unitary RNNs (Arjovsky et al., 2016), invented precisely to kill exploding and vanishing gradients.

The modern descendants are quietly everywhere: LRU and the S4/Mamba family use complex-diagonal recurrence with eigenvalues *near* the unit circle — mostly rotation with a whisper of decay. Pure tangential failed for an instructive reason: a system that never forgets can't prioritize. Memory needs erasure. The winning recipe is rotation for transport plus a learnable, input-dependent leak for forgetting.

And the physics reading is exact, not metaphorical: purely tangential dynamics is conservative — Hamiltonian flow, no dissipation. The decay term is dissipation. The empirical sweet spot is a *slightly dissipative* system riding just inside the conservative boundary.

> **Turning point:** "rotation plus a whisper of decay" becomes the conversation's central

> object. Everything after this is a different route to, or a stress test of, this same structure.

Turn 4 — Third probe: the linear part of the wave

Me: a fixed part of the tan wave is linear

Claude: Right — tan(x) ≈ x near zero, so a recurrence living in that band *is* linear recurrence. tanh has the same linear center, and that's the actual reason it won: a well-conditioned RNN spends most of its life in tanh's near-identity band, and the edge-of-chaos initialization literature formalizes tuning the Jacobian to sit there. So "a fixed part of the nonlinearity is linear, and we mostly stay there" isn't a what-if — it's a fair description of how trained recurrent nets already operate.

The difference is on *excursion*. tanh′ ≤ 1 everywhere: leave the band and you get compression — signals damp, the system forgets, stability guaranteed. $\tan' = 1 + \tan^2 \geq 1$ everywhere: leave the band and you get amplification accelerating toward actual poles at ±π/2. tanh is a dissipative wall; tan is an anti-wall.

But the charitable engineering reading: a system that is linear while normal and violently expansive when perturbed is not a memory — it's a **sensitivity amplifier**. An anomaly detector that stays quiet in the operating band and screams superlinearly the moment input pushes state toward the pole. You wouldn't train through it; you'd mount it as a fixed readout on a stable core — a nervous system's pain response bolted onto a calm memory.

> **Turning point:** tan is not rejected — it's *reassigned*. Wrong as a memory, plausible as an instrument. The conversation's habit of salvaging the probe instead of just correcting it starts here.

Turn 5 — Fourth probe: fix the derivative with coefficients

Me: tanh derivative won't be an issue for algorithms with coefficients

Claude: Coefficients help exactly once, and that's the catch. The recurrence Jacobian is diag(tanh′(h))·W, so yes — gain above 1 in W can cancel the average shrinkage; critical initialization is precisely that. The problem: tanh′ is a *state-dependent* multiplier, from 1 near zero to ~0.07 at |x| = 2, while your coefficients are fixed. Whatever gain you pick is correct at one operating point only. A W large enough to compensate in saturation has norm ~14 and detonates every unsaturated direction. Over T steps you multiply T of these mismatched products, and the errors compound exponentially — that's Pascanu's classic analysis, and it's an empirical wall, not a theory quibble.

Which is why every solution that worked made the coefficients *dynamic*: LSTM gates are input-dependent multiplicative coefficients whose whole job is holding the effective Jacobian at exactly 1 along the cell pathway — the constant error carousel. Mamba's input-dependent decay is the same move. Your instinct — fix the derivative with coefficients — is the right instinct; the field's finding is that the coefficients must be *functions of the state*. A fixed coefficient is a thermostat set once; the gate is a thermostat that reads the room.

> **Turning point:** the probe wasn't wrong, it was *static*. Promoting the coefficient from constant to function-of-state is the whole history of gating, compressed into one correction.

Turn 6 — Fifth probe: change the geometry

Me: if recurrence is based on trig and trig is the geometry of triangles defined in degrees and radians, then what to resolve issues of tanh′ and use proportional angles defined in radians

Claude: One correction that strengthens your argument: tanh isn't triangle trig at all — it's *hyperbolic*, the geometry of the hyperbola, not the circle. That's exactly why it saturates instead of cycling. So your proposal, read carefully, is: swap hyperbolic geometry for circular geometry — make the state an angle and update it by adding proportional radians. And that move genuinely dissolves the derivative problem: $d(\theta + \omega x)/d\theta = 1$, always, everywhere. Rotation has no saturation, no shrinking gradient, no pole — the only smooth update whose Jacobian is exactly norm-preserving by construction.

You've now derived, from the trig end, the same object we hit from the tangential end: state as phase, update as rotation, radians native. The mature versions are literally this — LRU/S4 eigenvalues are e^{iθ} accumulated in radians; RoPE encodes position by rotating dimension pairs through proportional angles; coRNN builds the network from coupled oscillators with proven bounded gradients. Phase-based memory is also the leading theory of biological neural state — phase survives where amplitude decays.

The price of the circle is the price of the circle: angles wrap. θ and $\theta + 2\pi$ are the same state, so pure phase stores direction but not magnitude, and a single frequency aliases. The fix is a *bank* of rotators at different proportional frequencies — fast dimensions for recent detail, slow for long context, which is precisely RoPE's spectrum — plus a small decay term so the system can forget. Rotation for transport, mild dissipation for forgetting, multiple frequencies for capacity: the complete recipe, every piece shipping in models today.

> **Turning point:** the hyperbola-vs-circle correction. My mislabeling of tanh as triangle trig, once fixed, made the radians proposal *stronger* — the correction was the unlock, not the

> rebuttal. Claude closed the turn with a line that forced the next one: "your chain of intuitions — tangential, then coefficients, then radians — walked the actual historical path in three turns."

Turn 7 — The challenge

Me: what do you mean by walked the historical path in three turns? what are you implying?

Claude: The plain version, both edges. Factually: your three moves each correspond to a real, named development — tangential → unitary/orthogonal RNNs; coefficients → gating; radians → phase/rotational state. What I'm implying positively: reconstructing that sequence from first principles is evidence your geometric intuition points where the field validated. That's true, not a dressed-up compliment.

What I'm *also* implying, and would be flattering you to omit: these ideas are not novel. Each was proposed, built, stress-tested, and absorbed years ago — the frontier is on the far side of them. If the unstated question is "am I onto a differentiated architecture," the honest answer is: not via this route. You're re-deriving the standard model, which is excellent education and poor differentiation. And a caution in the other direction: three correct intuitions in well-mapped territory is not evidence the fourth will be right where the map runs out. It's evidence you'd get real leverage from the primary sources — Arjovsky, Pascanu, the S4 papers — because you clearly metabolize this material, and the reading moves you from re-deriving known answers to standing where the open problems are.

> **Turning point:** the conversation's honesty register changes here. I asked what was being implied; the answer separated the compliment from the caveat instead of blending them. Everything after this turn is read against that standard.

Turn 8 — Into the primary source

I opened the S4 paper (arXiv:2111.00396) and hit the abstract's claim that the predecessor had "prohibitive computation and memory requirements."

Me: is this implying that it's considered prohibitive because compute and memory is over-consumptive or not enough?

Claude: It means too much — "prohibitive requirements" reads as "costs so high they prohibit using it." The specifics: the predecessor computed the SSM's convolution kernel by repeatedly multiplying by the state matrix A. The magic HiPPO matrix that gives long-range memory is *dense*, so that's roughly $O(N^2L)$ work with a memory footprint that blew up on real tasks. The obvious fix — diagonalize A so powers become elementwise — fails because HiPPO's diagonalization is numerically unstable. S4's whole contribution is the workaround: A isn't diagonalizable stably, but it *is* diagonal-plus-low- rank, and that structure reduces the kernel to a Cauchy kernel — near-linear. Same theory, tractable price.

And the connection to the thread you just walked: what does the diagonalized A look like? Complex eigenvalues near the unit circle — decaying rotations, angles in radians. The epilogue if you keep reading forward: Mamba discovered that once A is input-dependent you can drop the fancy structure for a plain diagonal. Dense → structured → diagonal-and-selective is the compute story of the whole SSM line.

Me: it didn't explicitly say too high

Claude: It did, in two pieces. "Prohibitive" isn't a neutral adjective awaiting a direction; it lexically means "so high as to prevent use" — nobody says "prohibitively little memory"; the construction doesn't exist. And the abstract closes the loop in the same sentence: "prohibitive computation and memory requirements, *rendering it infeasible*." A paper doesn't cure a resource shortage with an efficiency gain — it cures a resource *excess* with one. General convention worth banking: "prohibitive," "intractable," "infeasible," and "does not scale" are one family, and they always mean the cost curve grows faster than anyone can pay.

> **Turning point:** the conversation shifts from my proposals to a shared text. From here the probes are aimed at the paper's actual mechanism.

Turn 9 — Probing the mechanism: inversion

Me: i guess inversing won't make a difference either

Claude: Depends which inversion. If you mean inverting the *dynamics* — running A^{-1} — correct, no help, and the geometry says why instantly: a stable A has eigenvalues inside the unit circle; its inverse has them outside — rotation plus growth. You'd convert the dissipative system into an explosive one. Time-reversal of a dissipative system always looks like this — dissipation isn't invertible for free.

But if you mean matrix inversion as a *computational tool* — the joke's on the intuition, because inversion is precisely how S4 escaped the prohibitive cost. The kernel runs through the resolvent $(zI - A)^{-1}$, and the central trick is the Woodbury identity: the inverse of (diagonal + low-rank) is itself cheap and structured. Dense A made *iterating* expensive; structured *inverting* made it nearly free. The moral of the whole S4 line: no single operation is expensive or cheap in itself; cost lives in the *structure* of the matrix you apply it to.

Me: inversing the diagonal

Claude: Cheapest operation in linear algebra — reciprocal of each entry, O(N). The cost question disappears entirely. What survives is the dynamics question, and it collapses both halves into one statement: $|1/\lambda| = 1/|\lambda| > 1$ per coordinate — elementwise reciprocals still turn dissipation into explosion. Diagonalizing made inversion free; it didn't make the inverse *stable*. But as a computational primitive you've landed on something the pipeline genuinely uses: discretizing the continuous SSM computes $(I - \Delta A/2)^{-1}$, affordable exactly because diagonal inversion is elementwise — executed constantly inside S4 and Mamba. Sharpest form: **diagonal structure makes every operation cheap, but no representation change ever alters the eigenvalues, and the eigenvalues alone decide stable versus explosive. Cost is about structure; behavior is about spectrum. Independent axes.**

> **Turning point:** the two-axes principle gets stated for the first time. It becomes the frame the next three turns test, correct, and finally break.

Turn 10 — The ledger audit

Me: okay so this solves the prohibitive issues found in the ssm

Claude: Almost — with one arrow flipped. Cheap diagonal inversion isn't what solves the cost; it's a *consequence* of the thing that does, which is the diagonal parameterization itself. And the fine print: S4 couldn't use a pure diagonal — HiPPO resisted stable diagonalization — so its contribution was diagonal-*plus-low-rank* plus the Woodbury/Cauchy machinery. The satisfying epilogue: S4D and DSS later showed that discarding the low-rank correction barely changes performance — the field over-engineered around a constraint that turned out to be soft.

Me: right, we already went over that. you said though the issue was scale. i then brought up tan, then we did a linear part of tanh′, then coefficient function, now diagonal and the inverse

Claude: Close, but the ledger needs an audit, because two different problems got braided and your summary credits one thread with solving the other's issue. The conversation had two threads on two independent axes. Thread one — **stability**: tan, tanh′, coefficients, coefficient-functions, rotation in radians. That chain addresses whether gradients survive iteration. Thread two — **cost**: dense A prohibitive, diagonal elementwise, inversion riding along free. That chain addresses whether you can afford to compute it. The prohibitive issue was solved entirely by thread two — and your tan → coefficients sequence, good as it was, contributed nothing to it, because stability work can't reduce cost. They meet at exactly one joint: once A is diagonal, its entries turn out to *be* the rotation-with-decay elements from thread one.

I'd also flag the drift: the diagonal wasn't a step we derived — it's S4's published contribution, which I reported. The summary was quietly reorganizing the conversation into a narrative where your probes progressively cracked the SSM's cost problem, and that's not what happened. You built a genuinely good stability intuition, then read the cost story. Both are real; keeping them straight is what makes the next question land somewhere new instead of somewhere already mapped.

> **Turning point:** the audit. The conversation stops to correct its own emerging mythology. And the correction's closing sentence — "the next question lands somewhere new" — is immediately cashed in.

Turn 11 — The question that broke the frame

Me: naturally the next question is what if dense A and cost meet at a joint twice

Claude: Asked as a what-if, but it has a real answer — and it shows my "the two axes never interfere" line was one notch too clean. There is a second joint, and it's the exact reason HiPPO resisted diagonalization: **non-normality.**

Eigenvalues decide *asymptotic* behavior. But they only fully describe the dynamics when A is normal — eigenvectors orthogonal. Diagonal matrices are automatically normal. Dense matrices needn't be — and a non-normal A with every eigenvalue safely inside the unit circle can still produce large *transient* growth: trajectories that surge, sometimes enormously, before the decay wins. The spectrum says "stable"; the structure says "but first, amplification." That's structure shaping behavior through eigenvector geometry rather than eigenvalues — a second joint, orthogonal to the first. It's a mature field (pseudospectra — Trefethen), it's why HiPPO's diagonalizing transform was exponentially ill-conditioned, and it's dynamically *useful*: computational neuroscience models cortical responses as exactly this — brief selective surges from a stable system.

So the honest full statement: going diagonal for cost doesn't just reorganize the computation — it *forfeits non-normal transient dynamics*. The S4 lineage paid that price knowingly, and the follow-ups showed that for sequence benchmarks the forfeit was nearly free. But "nearly free on those benchmarks" is not "worthless" — in fluid dynamics and neural circuits, transient growth is the phenomenon of interest. This probe, unlike the earlier ones, lands on territory where the standard story genuinely has a wrinkle rather than a settled answer.

> **Turning point:** the payoff of the whole dialogue. After six probes that re-derived settled results, the seventh — asked right after the ledger was straightened — found the place

> where the clean principle actually bends. The audit wasn't pedantry; it was aim.

Turn 12 — Compression

Me: simplify

Claude: Two dials: how the matrix is *stored* (dense vs. diagonal) sets the cost, and its *eigenvalues* set whether it explodes or remembers. They connect once more: eigenvalues tell the whole story only when the matrix is "normal" — internal directions all at right angles. Diagonal always is. Dense needn't be — and a dense, non-normal matrix can be stable in the long run yet spike hard in the short run. Picture a guitar string guaranteed to go quiet eventually, but that twangs loudly first. The eigenvalues promise silence; the shape of the matrix produces the twang. Going diagonal removes the twang. Language benchmarks didn't miss it. In physics and neuroscience, the twang is often the signal itself. One sentence: **diagonal buys cheapness at the price of transient dynamics, and whether that price matters depends on whether your problem lives in the steady state or in the spike.**

Turn 13 — The pebble

Me: instead of eigenvalues getting more linear, the diagonal pushes towards more amplification like a pebble dropped in the water

Claude: Your metaphor is better than the mechanism you attached to it — let me swap the parts. What a diagonal *can't* do is push anything across dimensions: diagonal means zero coupling; each coordinate evolves alone, deaf to its neighbors. A ripple is propagation through a medium — it requires neighbor-to-neighbor coupling, which is exactly the off-diagonal structure. And a pebble's rings don't amplify: the splash is the loudest moment; the rings spread the same energy over an ever-larger circle.

But here's why your image is secretly the right one: **water waves are the textbook system that becomes diagonal in the right basis.** Decompose the pond into wavelengths — Fourier modes — and each mode evolves completely independently: its own frequency, its own phase speed, rotation-plus-decay, no coupling. The ripple you see in physical space is those independent rotating modes summed back together, interfering. The pebble is B — the input drop exciting every mode at once. The diagonal A spins each mode at its own rate. C reads the superposition back out. That's not an analogy to how S4 works — it's literally its mechanism, and its impulse response even *looks* like a decaying oscillatory ripple.

The corrected sentence: the diagonal doesn't push amplification outward like a pebble —

the diagonal is the pond *as seen per-wavelength*, and the ripple is what its independent rotations look like from outside. You reinvented the Fourier transform's reason for existing: coupling in physical space, freedom in mode space.

> **Turning point:** the pattern from Turn 4 completes. The wrong mechanism is stripped off the metaphor and the metaphor turns out to be the exact mechanism in a different basis. This is the image the piece you're reading exists to deliver.

Where it landed

Me: can this be drawn and is it worth sharing?

Claude: Yes to both. Three panels: the pebble and ripples in physical space (the input u); the pond decomposed into a stack of independent rotating-decaying dials with no arrows between them (the diagonal A made visible); the dials summed back into the ripple (C reading out the superposition). Caption: *"S4 doesn't simulate the pond — it simulates the dials."* Worth sharing as an explainer, not as research — the modal view of SSMs is known to the field, but the S4 literature is notoriously impenetrable, and nobody has done the visual version anchored in wave physics. Teaching, not claiming novelty.

What I take from it

Three of my intuitions — tangential motion, state-dependent coefficients, angles in radians — turned out to re-derive unitary RNNs, gating, and phase-based recurrence. That's validation of the compass and proof the territory was mapped. The two questions that went somewhere new were different in kind: one audited the conversation's own narrative before proceeding, and the next one — "what if they meet at a joint twice" — found non-normality, the real wrinkle in the clean story. The lesson I'm keeping: the probes only started landing on open territory after the ledger was honest about which probes hadn't. And the pond: coupling in physical space, freedom in mode space. The diagonal is the pond seen per-wavelength.

References for the curious: Arjovsky, Shah & Bengio, "Unitary Evolution Recurrent Neural Networks" (2016) · Pascanu, Mikolov & Bengio, "On the difficulty of training recurrent neural networks" (2013) · Gu, Goel & Ré, "Efficiently Modeling Long Sequences with Structured State Spaces" (S4, ICLR 2022) · Gu & Dao, "Mamba" (2023) · Trefethen & Embree, "Spectra and Pseudospectra" · Orvieto et al., "Resurrecting Recurrent Neural Networks for Long Sequences" (LRU, 2023) · Rusch & Mishra, "Coupled Oscillatory RNNs" (coRNN, 2021).